Surya Teja Alla

Data Analyst

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SUMMARY

Data Analyst with 6+ years of experience specializing in healthcare analytics, clinical outcomes, and population health management. Proficient in leveraging SQL, Python, Power BI, Tableau, and Snowflake to extract, transform, and visualize large-scale EHR, claims, and patient datasets for actionable insights and process optimization. Skilled in data modeling, predictive analytics, and ETL automation using Azure Data Factory, Databricks, and Alteryx, ensuring accuracy, scalability, and compliance with HIPAA and CMS guidelines. Collaborative team player with a strong ability to translate business requirements into analytical solutions, applying ICD-10, CPT, LOINC, and HEDIS standards to improve care quality, reduce readmissions, and enable data-driven decision-making across healthcare operations.

SKILLS

Programming & Scripting Languages: Python (Pandas, NumPy, Matplotlib, Scikit-learn), R (dplyr, ggplot2), SQL (T-SQL, PL/SQL), DAX, VBA, JSON, XML

Data Visualization & Business Intelligence Tools: Power BI, Tableau, Qlik Sense, Excel (Power Query, Pivot Charts, Macros), Looker Studio, Matplotlib, Seaborn

Databases & Data Warehousing:SQL Server, MySQL, PostgreSQL, Oracle SQL Developer, Snowflake, Azure Synapse Analytics, Google BigQuery, Teradata

Big Data Tools & ETL Tools: Azure Data Factory, Alteryx, SSIS, Informatica, Talend, Apache Airflow, Databricks, PySpark, API Integration,Pentaho , Apache NiFi , Apache Kafka , Apache Hive

Cloud Platforms & DevOps: Microsoft Azure, AWS (S3, Redshift, Glue), GCP (BigQuery), Docker, Kubernetes, GitHub, Azure DevOps, CI/CD Pipelines

Machine Learning: Supervised Learning (Regression & Classification), Unsupervised Learning, Ensemble Methods, Linear/Logistic Regression

Statistical & Analytical Techniques: Data Wrangling, Exploratory Data Analysis (EDA), Predictive Modeling, Statistical Inference, A/B Testing, Anomaly Detection, Forecasting, Feature Engineering

Healthcare Data & Compliance Tools:EHR/EMR Data Analytics, CMS Data Models, ICD-10, CPT, DRG, LOINC, HEDIS, HL7 Standards, HIPAA Compliance, Population Health Analytics, Claims Processing Systems, Clinical Quality Metrics

Version Control & DevOps: Git, GitHub, Docker, Jenkins, CI/CD Pipelines

Project Management & Collaboration Tools: JIRA, Confluence, MS Visio, Lucidchart, Agile/Scrum Methodology, Power Automate, ServiceNow, Slack, Trello

EXPERIENCE

United Healthcare Group, USA | Sr. Data Analyst Mar 2025 – Present

- Architected an enterprise-grade data integration framework using SQL Server, Snowflake, and Azure Data Factory to unify EHR, claims, and pharmacy data mapped with ICD-10 and CPT standards, improving data quality by 35% and accelerating clinical data validation by 60% across care networks.
- Developed predictive readmission models with Python (Pandas, Scikit-learn) and Databricks, performing advanced feature engineering and risk stratification to identify high-risk chronic patients, which increased model precision by 25% and optimized care team targeting for early intervention.
- Implemented automated ETL pipelines and workflow orchestration in Azure Data Factory and containerized deployments via Docker and Kubernetes, reducing data latency by 85% and enabling near real-time analytics for population health management dashboards.
- Designed and deployed interactive Power BI dashboards integrated with CI/CD pipelines on GitHub and Azure DevOps, enhancing visualization of readmission trends, provider efficiency, and cost-of-care KPIs, leading to a 70% faster decision-making cycle for healthcare executives.
- Collaborated with clinicians, actuaries, and data engineers to translate predictive insights into actionable strategies using Tableau, Apache Hadoop, HEDIS, and CMS Star Ratings, improving chronic disease management compliance by 40% and elevating overall quality performance across multiple provider groups.

Happiest Minds Technologies, India | Data Analyst May 2019 – Apr 2023

- Consolidated payer, provider, and EHR datasets through SQL Server, Snowflake, and Azure Synapse while integrating ICD-10, CPT, and DRG taxonomies which enhanced claims traceability by 30% and improved data governance consistency across multiple healthcare networks.
- Engineered predictive fraud and anomaly detection workflows using Python, PySpark, Talend, and Databricks, performing cluster-based analysis and feature extraction that increased fraud identification precision by 25% and accelerated claim adjudication in compliance-sensitive environments.
- Designed population health dashboards in Tableau and Power BI integrated with Docker and Kubernetes orchestration for real-time data visualization which improved report accessibility by 70% and enabled executives to monitor chronic disease KPIs seamlessly.
- Automated ETL and metric computation pipelines in Azure Data Factory, AWS (S3, Redshift, Glue), and Power Query combined with CI/CD integration via GitHub and JIRA workflows, increasing data refresh efficiency by 40% and supporting continuous quality monitoring aligned with LOINC and HEDIS healthcare data standards.
- Collaborated with actuaries, clinicians, and business teams using Excel, API Integration, and Advanced SQL to generate predictive health risk scores and reimbursement insights achieving a 35% uplift in preventive care performance and ensuring HIPAA-compliant analytics delivery.

Persistent System, India | Business Data Analyst Jul 2017 – Apr 2019

- Strategized analytical frameworks using SQL Server, Oracle SQL Developer, and Azure Synapse to merge patient admission, discharge, and billing data mapped with ICD-10 and CMS readmission standards improving operational data transparency by 30% and enabling proactive patient outcome tracking across multi-facility hospitals.
- Translated business and clinical requirements into technical workflows through Power BI, MS Visio, and Lucidchart, defining KPIs such as readmission rate and discharge efficiency which strengthened decision-making by 25% and aligned analytical outputs with hospital performance goals.
- Evaluated revenue cycle inefficiencies by integrating payer and reimbursement datasets using MySQL, Alteryx, and CPT-based classification models uncovering payment delays and claim denials that enhanced billing accuracy by 20% and supported cost recovery for healthcare finance teams.
- Optimized financial and clinical dashboards in Tableau, Power BI, and AWS, automated through RPA with Confluence documentation, reducing manual data processing by 40% and delivering real-time insights on readmission trends and payer turnaround times..
- Collaborated cross-functionally with physicians and engineering teams in Agile sprints using Jira, applying data modeling, variance analysis, and forecasting techniques that improved cross-departmental visibility by 35% and ensured compliance with HIPAA and CMS standards for healthcare data governance.

EDUCATION

Master of Science Technology Management — Lindsey Wilson College, 210 Lindsey Wilson Street, Columbia, USA May 2025